

A Multi-Sensory Music Remote



: Enabling Autonomy
on The Possibilities
Floor

Abstract

Residents on Baycrest's Possibilities Floor experience varying levels of meaningful engagement in their day-to-day activities. They require constant staff cueing, which can be demeaning for residents and exhausting for Mavens. Existing recreational tools fail to hold resident attention, as they are often too passive or require fine motor skills that many residents lack.

1 Introduction

The Possibilities Floor serves residents with varying cognitive abilities, but activities are mostly one-size-fits-all. During our time there, we noticed a consistent pattern: Mavens constantly prompt residents to participate. For residents, this can feel controlling and/or infantilizing; for staff, it can be exhausting.



If we can give residents the ability to control light, sound, and vibration by resting their hand on different textures, we can shift them to a more active level of engagement with more autonomy and control.



2 Research Methods

We chose these methods because seeing residents and staff in their actual context was crucial. We needed to understand not just what they said they wanted, but what they actually engaged with. The Mavens in particular had a wealth of knowledge and some great ideas, so the direct interviews were essential.

Observational
shadowing on the
Possibilities Floor
(multiple sessions)

4 structured
interviews with
Mavens

Empathy mapping
for both
residents and
staff

Tracked how residents
actually responded to
dynamic elements already
at Baycrest (the colored
pucks, projector table)



3 Key Insights

Residents naturally gravitate toward moving or dynamic elements, like the hallway pucks, without needing to be told to interact. Real engagement occurs when the environment itself sparks curiosity. Research supports this: well-designed sensory spaces lead to more independent engagement and less staff prompting (SonderCare)

The Environment Itself Prompts Engagement

Static Activities Get Old Quick; Change Keeps Residents Interested

Personal Hobbies Are Overlooked, But They Matter

Sensory Cues Beat Verbal Instructions, Especially for Nonverbal Residents

Higher-Functioning Residents Need a Personal Touch

Music Works When Words Don't

Constant Staff Prompting Isn't Working

"How Might We" Questions

- How might we help residents feel a sense of autonomy through multisensory feedback (light, sound, vibration) in order to help increase a feeling of independence and reduce Mavens' workload?
- How might we create a multisensory remote that keeps residents engaged on their own in order to reduce Mavens' workload?



Usability Testing 4



We tested our rough prototype with two groups: the Residents (our users) and the Mavens (the experts/caregivers). Our goal was to see how they interacted with the Sonic Touch with minimal instruction. We handed them the prototype, observed how they held it, and listened to their immediate feedback on its feel and function.

Key Questions:

- "Does the device feel comfortable to hold?"
- "Where would you naturally touch this to make it work?"
- "Is the feedback you get from the device satisfying?"
- "Is this something you would enjoy using?"

Key Findings

The Issue:

Residents struggled to find the specific touchpoints on their own. They often needed us to guide their hands to get a reaction from the device.

The Fix:

This indicated that the "buttons" weren't obvious enough. Our main goal for the final refinement was to make the interactions clearer, ensuring Residents can use the Sonic Touch independently without needing someone to show them how.



5 Market Research

When we examined the current market for dementia care products, we found that existing options fall into two extreme categories:



Passive Fidgets:

These include fidget muffs, therapy dolls, and blankets. They are portable and inexpensive, but they don't provide much engagement.



Big Systems:

These include Snoozelen rooms and interactive projectors. They offer great feedback and stimulation, but they are very expensive, require a lot of space, and usually need a caregiver for setup.

The Gap:

We identified a missing link in the middle. There is a need for a device that is portable, like a fidget, but also responsive. It should be simple enough for lower-functioning users to hold and to provide feedback.

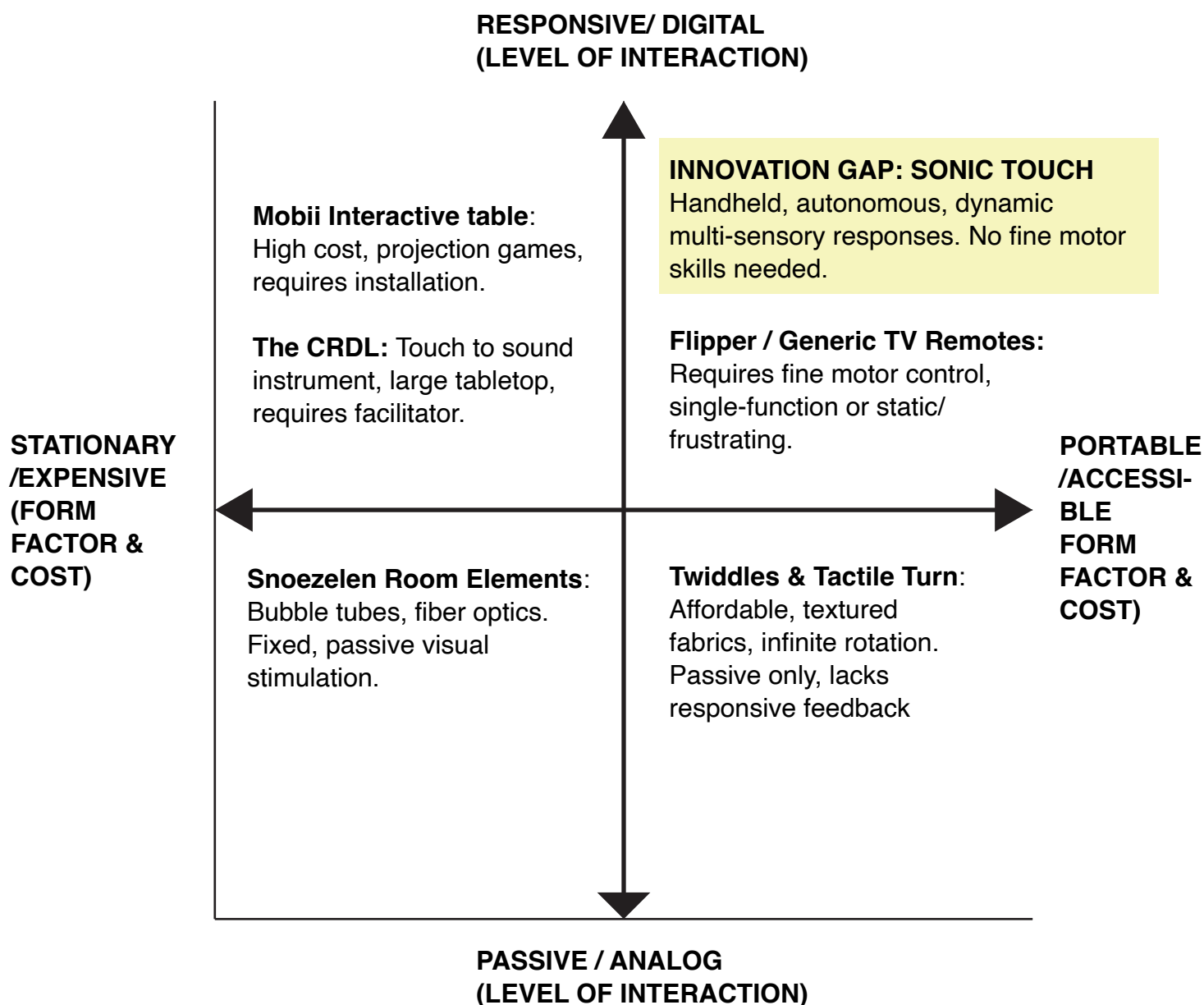
Competitor Highlight: The CRDL

- An interactive instrument that turns touch into sound. An example of technology that works well for therapy and connection.
- CRDL is large and used as a tabletop object designed for two people (a resident and a caregiver)
- Our approach: take the successful "touch-to-sound" idea and make it autonomous. Unlike the CRDL, our device is handheld. A resident can pick it up and use it independently, wherever they are, without needing a caregiver to facilitate the session.



6 Research Market Matrix

PRODUCT INNOVATION MATRIX: SENSORY DEVICES FOR DEMENTIA CARE



7 Key Takeaways

How It Solves the Problem:

Direct Control: Touching a particular fabric leads to an immediate response. There are no buttons to find and no fine motor precision is required. The causality is obvious: I did this, and that happened.

Novelty: Each interaction feels slightly different (different LED colors, varying sound samples, different vibration patterns). Residents stay curious instead of bored.



Multisensory Experience: Light + sound + vibration together hold users' attention longer than single modes do, as research supports (Mileski et al.).

Reduced Prompting: Without this device, Mavens spend more time prompting residents. With it, residents can engage independently, without waiting to be prompted.

What It Doesn't Solve:

Social connection: The device is individual-focused. It doesn't create peer interaction or replace human conversation.

Complex engagement: Higher-functioning residents might find it under-stimulating. They might prefer games, choice boards, or personalized activities.

Long-term novelty: After months of use, some residents might habituate (get used to the novelty). We didn't test for that.

Personalization: The device responds the same way for all residents. Adding recorded family voices or personalized music could deepen meaning but adds complexity.

Physical barriers: If a resident is in pain or acutely agitated, they won't engage no matter how good the design is.



8 Future Work

Implementation Considerations

Durability & Hygiene: In a care home, drops and spills are inevitable. Manufacturing must utilize medical-grade, non-porous materials, like silicone overmolding, which can withstand impact and constant sanitization.

Customization vs. Simplicity: One size doesn't fit all. To ensure vibration soothes rather than startles, we recommend a hidden "caregiver menu." This allows staff to adjust intensity without cluttering the resident's simple interface.

Power Management: True autonomy means no cables. A wireless charging dock (similar to hearing aids) is essential so residents don't have to fiddle with cords, ensuring the device is always charged and ready to grab.



Future Research

Long-term Engagement: Does the novelty wear off? Future tracking over weeks or months is needed to determine if the software requires "evolving" feedback patterns to maintain interest.

Therapeutic Efficacy: We know they like it, but does it work? A clinical pilot study tracking behavioral incidents is required to prove if the device measurably lowers anxiety or the need for medication.

Thresholds of Sensory Overload: How much is too much? We need to investigate the fine line between stimulation and overstimulation to develop scientifically-backed "Calm" versus "Active" presets.



9 Product Architecture

The Device:

Lap-based & handheld, roughly 10" x 8" x 3"

Divided into four colored sections, each with their own capacitive touch sensor and one small capacitive touch sensor in the middle. The 4 sensors play gentle harmonizing tones, vibrate and illuminate a corresponding LED. The middle button can cycle through a handful of different relaxing tones. The sensors are also pressure sensitive and will raise intensity of the sensory response with the pressure. There is an internal volume and sensory mode button to cycle through any combination of sensory responses (Sound, Vibration, Light).

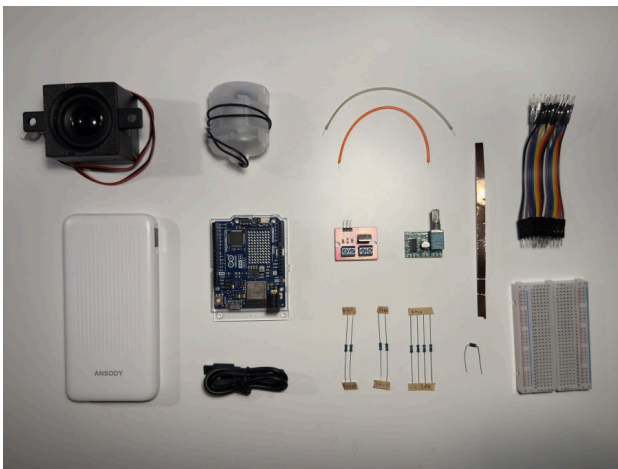


Why This Design:

Capacitive sensors respond to any touch; open hand, partial hand, or just resting. No precision is required. Textured fabrics are visually distinct and invite exploration. Multiple fabric zones mean multiple response profiles, maintaining novelty. Hidden electronics keep the focus on tactile experience, not buttons and controls.

Electronics:

- Arduino Uno (or Nano for size)
- 4-6 capacitive touch sensors (one per zone, plus redundancy)
- 4 RGB LEDs (or 4 individual colored LEDs) mounted internally behind translucent fabric
- 1 vibration motor (24mm, 12V)
- Speaker
- 10,000mAh portable power bank
- Mode switch (inside): Full Multisensory / Music-Only / Light-and-Vibration



10 Human Factors in Design



Affordance: The fabrics afford touching and invite tactile exploration without any instruction. Unlike hard plastic buttons, texture communicates "touch me" immediately.

Signifier: The immediate multisensory response (light + sound + vibration) signals causality. Resident touches fabric -> immediate feedback. They understand: I caused this.

Mental Model: Residents retain basic cause-and-effect understanding even in advanced dementia. We aligned the device with this: simple cause (touch), immediate effect (sensory response). No learning curve needed.

ACCESSIBILITY DETAILS:

- No fine motor requirement (capacitive sensors are forgiving)
- High-contrast, bright LEDs (visible despite vision loss)
- Clear, non-muddled audio (compensates for hearing loss)



11 Design Development Process

1. Product Architecture & Proposed Response

The Sonic Touch is a handheld, autonomous sensory device designed to bridge the gap between passive fidgets and complex sensory rooms. It is a direct response to the "innovation gap" we identified between low-engagement items and high-cost systems.

The Human Challenge: At Baycrest, we saw that the environment was "hyper-monitored" but didn't actually let residents do much on their own. They were often stuck waiting for Mavens (PSWs) to prompt them. When tech like the Nintendo Switch was tried, it failed because it needed fine motor skills that residents just didn't have. We also noticed a "screen stigma": families viewed residents watching TV as "passive," whereas tactile engagement (like a resident playing with a necklace) was seen as positive.

The Proposed Response: The Sonic Touch prioritizes autonomy and dignity. It's a "pick-up-and-play" device with no setup and no screens.



- Housing: A hand-turned solid cherry bowl. We picked wood to avoid the "medical" look of plastic, aligning with research that says home-like objects reduce anxiety (SonderCare).
- Base: A custom bean bag cushion that attaches magnetically to the bottom. This lets it mold to a lap or uneven surface, which is huge for residents in wheelchairs.
- Sensory Inputs: Copper tape capacitive sensors exposed through circular cutouts in felt patches.
- Internal Engine: It's powered by a complex circuit (Arduinos, MOSFET, 12V motor) that drives the vibration, sound, and LEDs.

How Testing Changed the Design: Usability testing was the deciding factor for the final interface. Our early prototype hid the sensors to look sleek, but it failed in testing. The residents liked the feedback but didn't know where to touch without help. This killed the autonomy we were aiming for. So, we changed the architecture to use felt patches with holes cut in the center. This frames the copper sensor, creating a clear "target" that tells the user exactly where to place their finger without a staff member stepping in.



2. Embedded Human Factors in Design

We used Norman's interaction principles to make sure the device is intuitive and solves the "over-prompting" issue we saw at Baycrest.

Affordances (The Shape): The device uses the natural affordance of a bowl. The curved, tapered wooden form fits naturally in two hands, inviting you to hold it. This supports findings by Cruz et al. that interventions should be "motor-based" and accessible (Cruz et al. Making Sense(s)). Unlike a remote control that you point, the Sonic Touch affords cradling, which is grounding.

Signifiers (The Clues): To fix the "guidance" problem, we used a distinct material contrast:

VISUAL

Visual: The bright felt patch contrasts against the dark wood, and the shiny copper contrasts against the matte felt.

TACTILE

Tactile: The hole in the felt creates a physical guide. The finger naturally slides from the soft felt into the "well" of the cool copper tape.

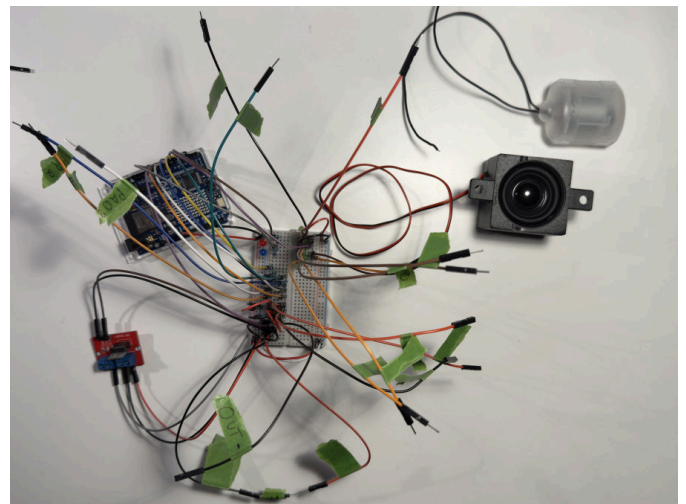
FEEDBACK

Feedback: As soon as you touch the copper, you get immediate confirmation (Light + Sound + Vibration). Mileski et al. note that this kind of responsive feedback is key to keeping people engaged in long-term care (Mileski et al.).

3. High-Fidelity Prototype

Our final model is a fully functional "looks-like, works-like" prototype.

- **Materiality:** The shell is solid cherry, turned on a lathe and finished smooth. It feels valuable and warm, not cold and clinical.
- **Interface:** The interaction points are defined by industrial felt patches with a central aperture that exposes the copper Tape.
- **Functionality:** It's not just a model; it works. Touching the exposed copper completes the circuit, activating a programmed sequence of calming tones, soft LED illumination, and deep haptic vibration (via the 12V motor). This shows that a portable, battery-operated multi-sensory environment is feasible.

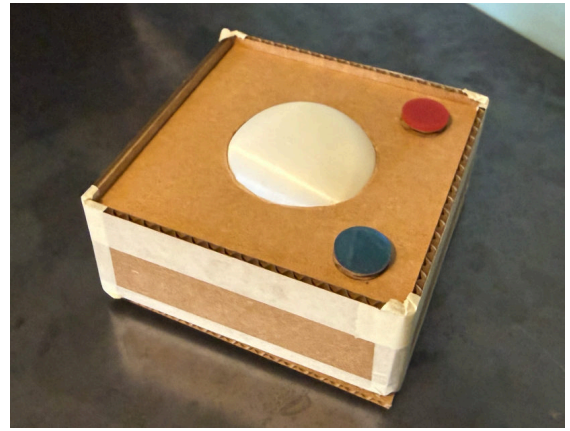


4. Design Development Process (Iterations)

Iteration 1: Form Finding (Cardboard & Foam)

Research Insight: Watching residents struggle with Switch controllers showed us we needed gross motor accessibility, not fine motor precision.

Physical Output: We made foam models to test the scale. We ditched "controller" grips for organic, rounded shapes that could be held with a flat palm or closed fist.



Iteration 2: The "Works-Like" Rig (Functional Cardboard Box)

Research Insight: Research says meaningful engagement needs dynamic feedback (Maneemai et al.). We had to prove a small device could actually produce enough output to be felt.

Physical Output: We built a rough "ugly" box to house the wiring.

Key Learning: The vibration motor rattled against the hard casing, making an annoying noise. This taught us the final housing needed mass (solid wood) to dampen the sound.



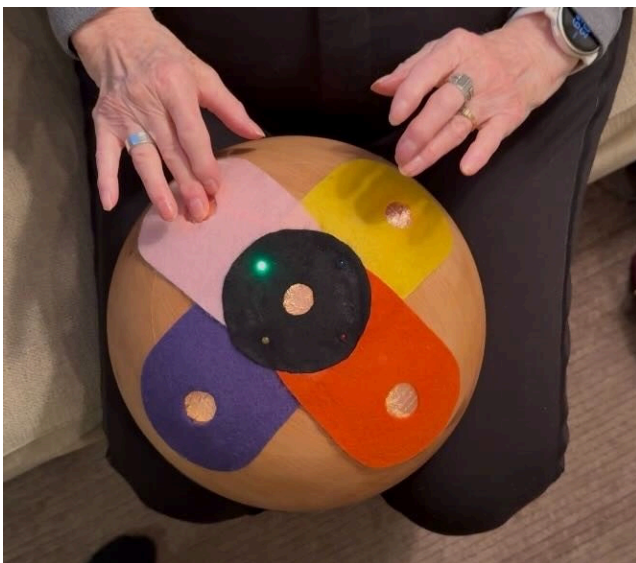
Iteration 3: The Sonic Touch (Final Integration)

Research Insight: Testing showed the form was good, but the "hidden" tech was too subtle. Residents needed clear cues.

Physical Output: We lathed the final wooden housing and implemented the Felt-Framed Copper Sensors. This combined the robust electronics of Iteration 2 with the ergonomic form of Iteration 1, finally solving the "autonomy gap."



12 Storyboard of Product in Use



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